How Complete Should a Reference Be? A Benchmark Audit for Fluorescence Spot Detection

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Abstract

Fluorescence spot-detection benchmarks in biology and medicine often rely on reference annotations that are incomplete, ambiguous, or inconsistently defined. In such settings, detector performance is not determined solely by model predictions; it can also depend on which faint or ambiguous signals are counted as reference positives. We formulate this issue as a *reference-audit problem*: instead of treating the evaluation reference as fixed ground truth, we test whether benchmark outcomes remain stable under plausible changes in reference definition and annotation completeness. Using the public FISH_spots dataset, a benchmark for fluorescence in situ hybridization (FISH) spot detection, we construct Raw, Lenient, and Strict audit references and evaluate diverse classical and learning-based detectors under a unified point-matching protocol. We further apply annotation-retention stress tests that simulate random and structured missing-label mechanisms. Across these audits, changing only the evaluation reference can alter measured F1 scores and model rankings, including the selected top-1 detector, even when aggregate rank correlation remains high. These findings suggest that model selection in fluorescence spot-detection benchmarks should be tested across plausible reference definitions, rather than based on a single reference annotation set.

1 Introduction

Fluorescence spot detection aims to locate small, discrete fluorescent signals ("spots") in microscopy images so that they can be counted and analyzed. Recent learning-based spot detectors, including deepBlink [3], U-FISH [17], and Spotiflow [2], have improved detection performance across diverse imaging conditions. Yet reliable evaluation remains difficult. Microscopy images often contain dense clusters of tiny, low-contrast, and visually ambiguous spots, making it hard to annotate every plausible signal completely and consistently [11, 17]. Even with expert inspection, faint spots may be missed, clustered signals may be inconsistently separated, and biomedical image annotations can exhibit inter- and intra-observer variability [4, 14].

This creates a benchmark-level problem. Most spot-detection benchmarks report detector performance against a single fixed reference annotation set. However, when that reference contains missing positives, ambiguous positives, or implicit annotation criteria, the measured score reflects not only detector quality but also the assumptions embedded in the reference itself [7]. The issue is especially consequential for small and tiny objects, where a single missing annotation or a small localization shift can affect point matching, precision, recall, and F1 estimates [1, 15]. Thus, in fluorescence spot detection, the reference annotation should be treated not as a passive measurement target, but as part of the evaluation protocol.

Figure 1 illustrates the reference-definition problem studied in this paper. The same microscopy image can induce different evaluation targets depending on how reference positives are defined. A

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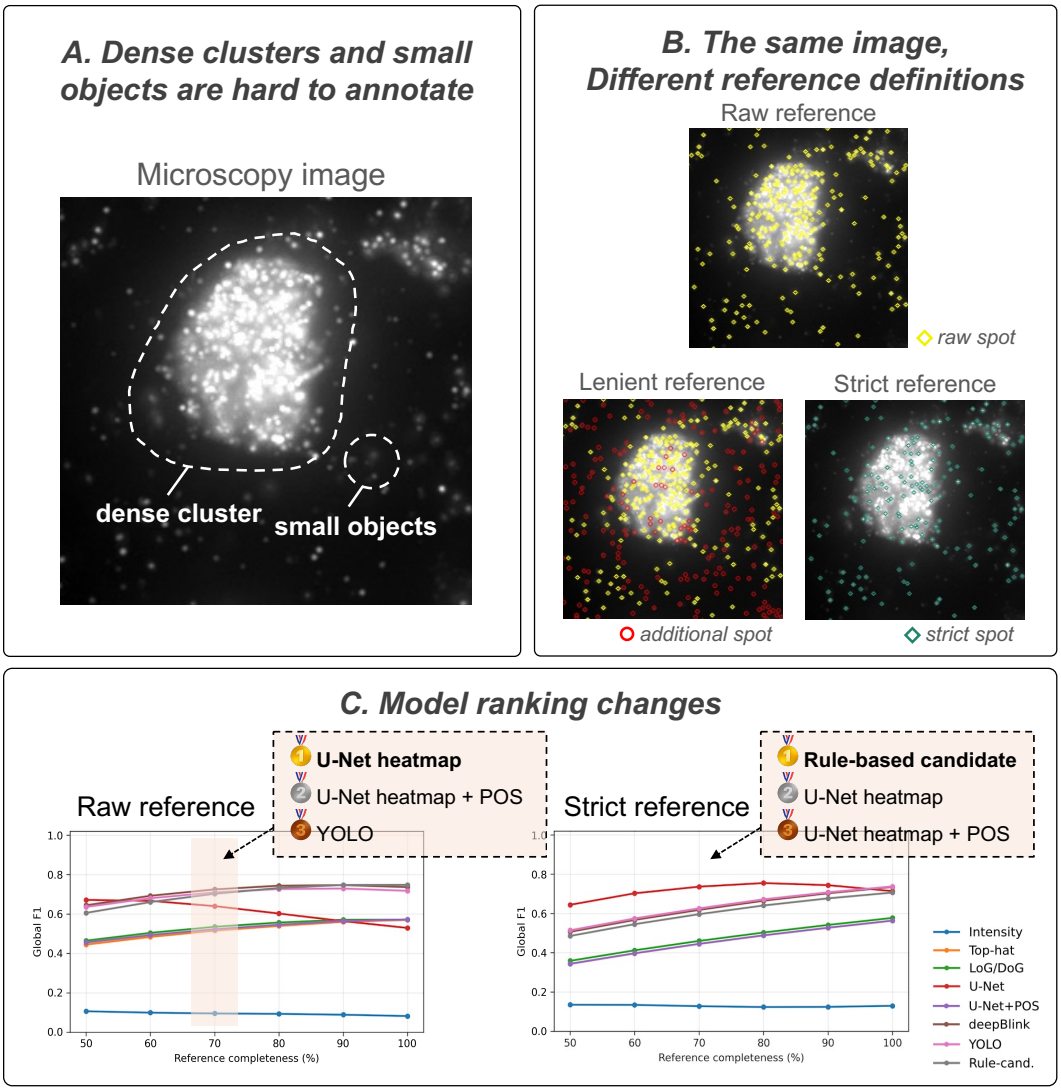


Figure 1: **Fluorescence spot-detection benchmark conclusions can depend on how the evaluation reference is defined. A:** Dense clusters and small objects can contain faint, crowded, and visually ambiguous spots, making complete and consistent annotation difficult. **B:** The same image can define different evaluation targets under Raw, Lenient, and Strict audit references. **C:** F1 scores of eight detectors as reference completeness decreases from 100% to 50%, evaluated separately under the Raw and Strict reference views. Each curve represents one detector, and the ranking boxes highlight the top three detectors at the indicated completeness levels.

conservative reference may retain only high-confidence spots, whereas a more inclusive reference may also include candidate-supported plausible spots. As a result, the same detector predictions can receive different measured scores, and the resulting benchmark ranking can change even when the images and predictions are fixed.

We study fluorescence spot detection as a reference-audit problem rather than a model-design problem. We use the public FISH_spots dataset, a benchmark for fluorescence in situ hybridization (FISH) spot detection [17, 16], as a testbed and construct Raw, Lenient, and Strict reference definitions that encode different assumptions about annotation completeness and ambiguity. We do not treat any reference as absolute biological ground truth. Instead, we use these references as controlled evaluation views for measuring how detector scores and rankings respond to changes in the evaluation target.

We evaluate diverse classical and learning-based detectors under a unified point-matching protocol and reuse the same stored scored predictions across reference definitions. Model training, image sampling, and matching criteria remain unchanged, while evaluation cutoffs follow the procedures described in Appendix B. Beyond standard evaluation, we define four annotation-retention stress tests that simulate plausible sources of missing labels during reference construction. These include random omissions and structured omissions of weak-evidence spots, crowded-region spots, and boundary-adjacent ambiguous spots.

On the evaluated FISH_spots subsets, our analysis shows that reference construction can substantially affect both absolute performance estimates and model-selection outcomes. In particular, aggregate rank-correlation metrics can remain relatively stable even when the selected top detector changes under incomplete references. This distinction matters because benchmark use often depends on choosing the best-performing method, not only on preserving the full ordering of all methods.

In summary, our contributions are as follows:

- We formulate fluorescence spot-detection benchmarking as a *reference-audit problem*, where the evaluation reference is treated as an explicit variable rather than fixed ground truth.
- We construct Raw, Lenient, and Strict reference views for the public FISH_spots dataset and provide a unified audit protocol for evaluating detector sensitivity to reference definition.
- We introduce annotation-retention stress tests that simulate random and structured forms of reference incompleteness, including weak-evidence, crowded-region, and boundary-adjacent omissions.
- We show that reference incompleteness can change measured F1 scores, detector rankings, and top-1 model selection, even when aggregate rank correlation appears stable.

2 Related work

Fluorescence spot detection. Fluorescence spot detection is a core step in image-based spatial omics, transcript localization, and subcellular structure quantification. Classical pipelines commonly use local intensity maxima, morphological filtering, and scale-space blob detectors such as Laplacian-of-Gaussian/Difference-of-Gaussian (LoG/DoG)-based methods [13, 12, 5, 6]. Recent learning-based detectors, including deepBlink [3], U-FISH [17], and Spotiflow [2], have improved spot localization across microscopy settings. These methods are usually evaluated against a fixed reference annotation set, whereas our work studies how benchmark conclusions change when that reference itself is varied.

Annotation noise and incomplete references. Noisy or incomplete annotations can distort object-detection evaluation by introducing missing positives, inaccurate localization, and inconsistent treatment of ambiguous instances [7, 14]. This issue is especially important for small and tiny targets, where a single missing annotation or small localization mismatch can affect precision, recall, and F1 estimates [1, 15]. In fluorescence spot detection, missingness may also be structured: faint spots, crowded spots, and boundary-adjacent spots may be more likely to be omitted than visually obvious isolated spots. Our work therefore evaluates not only annotation incompleteness, but also the missingness mechanism.

Benchmark reliability and evaluation auditing. Benchmark results shape model selection, but prior work has shown that label errors and test-set construction can destabilize reported rankings and generalization conclusions [7, 8]. Rather than treating the reference as fixed ground truth, we formulate fluorescence spot-detection benchmarking as a reference-audit problem. We construct Raw, Lenient, and Strict reference definitions and apply structured missingness stress tests to quantify whether detector scores, rankings, and top-model selection remain stable under plausible changes in reference construction.

3 Benchmark audit framework

3.1 Dataset and audit subsets

We use the public FISH_spots dataset [17, 16], which contains 2D fluorescence microscopy images with point-level spot annotations. We selected MERFISH, RCA, and SeqFISH because they contain

real experimental images with contrasting spot density, crowding, signal appearance, and background characteristics.

Table 1 summarizes the dataset split sizes; the audit-set refinement procedure is detailed in Appendix A.

The audit test set contains 50 images from each subset, balancing image counts across subsets of different sizes. Starting from random samples drawn with seed 42, we separated five sparse/near-empty images per subset into a control set and sampled replacements from the same subset. The final audit contains 150 images.

Before evaluation, all annotations and detector predictions are standardized into a common point-based format.

Table 1: FISH_spots dataset split.

Subset	Train	Val	Test
deepBlink	570	63	–
DeepSpot	450	50	–
ExSeq	432	48	–
MERFISH	450	50	50
RCA	902	100	50
SeqFISH	361	40	50
simfish	450	50	–
Total	3615	401	150

3.2 Reference views

For each image index i, let R_i denote the original Raw reference point set and C_i the candidate point set generated from image evidence. Candidate evidence is used to construct alternative reference views for auditing; it is not treated as an oracle or as definitive biological ground truth.

Let $d(r, c) = \|r - c\|_2$ denote Euclidean distance in pixels, and let τ_{ref} denote the reference-construction tolerance, distinct from the detector-evaluation matching tolerance τ_{eval}. We define candidate support and additional candidate points as follows:

$$s_i(r) = \mathbb{I}\left[\exists c \in C_i : d(r, c) \le \tau_{\text{ref}}\right], \quad r \in R_i,$$
$$A_i = \{c \in C_i : d(c, r) > \tau_{\text{ref}} \text{ for all } r \in R_i\}.$$

Here, $\mathbb{I}[\cdot]$ is the indicator function, equal to 1 when the enclosed condition holds and 0 otherwise. Thus, $s_i(r) = 1$ when at least one candidate lies within τ_{ref} of r, while A_i contains candidates with no Raw point within this tolerance.

We define the three audit reference views as follows:

$$R_i^v = \begin{cases} R_i, & v = \text{Raw}, \\ R_i \cup A_i, & v = \text{Lenient}, \\ \{r \in R_i : s_i(r) = 1\}, & v = \text{Strict}. \end{cases} \tag{1}$$

Table 2: Reference definitions used for audit-based evaluation.

Reference	Operational definition	Interpretation
Raw	Original annotations provided by the dataset.	Baseline benchmark reference.
Lenient	Union of Raw annotations and audit-accepted additional candidate spots.	Permissive audit view including additional plausible positives.
Strict	Raw annotations supported by candidate evidence.	Conservative high-confidence reference.

Raw, Lenient, and Strict are not intended to represent definitive biological ground truth. Instead, they define complementary audit views that allow us to test whether benchmark scores and model rankings remain stable under plausible changes in reference construction.

Candidate points are generated from image-level signal evidence using local peak structure, background-corrected intensity response, a signal-to-noise ratio (SNR) proxy, and a Laplacian-of-Gaussian (LoG) response score. For each subset, we fix the candidate-generation and audit-filtering parameters before final detector evaluation. Candidate points are filtered using an SNR threshold and a subset-specific LoG-response percentile threshold, and a candidate cap is applied to prevent images with excessive candidate counts from dominating the audit reference.

Table 3: Candidate-generation and audit-filtering settings used to construct Lenient and Strict reference views.

Subset	τ_{ref}	Evidence score	Candidate filtering threshold	Candidate cap
MERFISH	3 px	SNR proxy / LoG response	SNR $\geq$ 3.0, LoG response $\geq$ 95th percentile	3000
RCA	3 px	SNR proxy / LoG response	SNR $\geq$ 3.0, LoG response $\geq$ 96th percentile	2000
SeqFISH	3 px	SNR proxy / LoG response	SNR $\geq$ 3.0, LoG response $\geq$ 95th percentile	3000

These parameters are fixed for reference construction and are not tuned to maximize the test-set F1 score of any evaluated detector. Table 3 summarizes the subset-specific construction settings.

Table 4 reports the number of microscopy images and reference points in each audit reference view. The same 150 images are used across all three views, while the number of reference points differs according to the reference definition.

Table 4: Reference-audit test-set sizes. Images counts microscopy images; Raw, Lenient, and Strict count reference points.

Subset	Images	Raw	Lenient	Strict
MERFISH	50	40,362	48,844	32,546
RCA	50	23,420	31,958	13,731
SeqFISH	50	16,092	20,348	11,131
Total	150	79,874	101,150	57,408

3.3 Detectors and point-matching evaluation

We evaluate eight detectors spanning classical image-processing baselines, learning-based spot detectors, an object-detection baseline, and a rule-based candidate generator. The detector pool is not intended to introduce a new state-of-the-art method; it is used to test whether reference-dependent conclusions persist across heterogeneous detection strategies.

Table 5: Detectors evaluated in the benchmark audit. All outputs are standardized into point coordinates before evaluation.

Detector	Category	Standardized output
Intensity peak [13]	Classical	Local-maxima points
Top-hat peak [12]	Classical	Background-corrected peak points
LoG/DoG [5, 6]	Classical	Scale-space blob/peak points
U-Net heatmap [10]	Learning-based	Heatmap peak points
U-Net heatmap-pos [10]	Learning-based	Positive-patch heatmap peak points
deepBlink [3]	Learning-based	Point predictions
YOLO point-box [9]	Object detection	Bounding-box centers
Rule-candidate	Rule-based	Candidate points

For point detectors, the predicted coordinate is used directly. For heatmap-based detectors, local maxima after post-processing are converted into point coordinates. For box-based detectors, the center of each predicted bounding box is used as the predicted point. Detector checkpoints, prediction-generation settings, and stored scored predictions were fixed across reference views. Evaluation cutoffs were selected separately using audit F1, with analysis-specific procedures detailed in Appendix B. In reference-invariant cutoff controls excluding Rule-candidate, both Raw-selected and pooled rules selected YOLO under Raw and Lenient and U-Net heatmap under Strict. Rule-candidate exclusion results are reported in Table 7, with additional details in Appendix H.1.

For detector m, let $P_{i,m}$ denote the predicted point set for image I_i. Given an audit reference R_i^v, predictions and reference points are matched using one-to-one point assignment within a fixed spatial tolerance τ_{eval}. Unless otherwise stated, we use $\tau_{\mathrm{eval}} = 3$ pixels. Unmatched predictions are counted as false positives, and unmatched reference points are counted as false negatives. We aggregate counts over the held-out audit images and report precision, recall, and F1 score. F1 is used as the primary operational metric for comparing detectors under each fixed reference view. Detailed matching rules and metric definitions are provided in Appendix C. Data access, released audit inputs, and reproduction instructions are described in Appendices L and M.

3.4 Structured missingness and stability metrics

Table 8 in Appendix B summarizes the fixed and varied components of the audit.

To test whether benchmark conclusions remain stable under incomplete references, we construct degraded references by retaining only a fraction α of each reference view. Within each reference view and subset, the prediction set used for evaluation is held fixed as F1 is recomputed after removing reference points according to one of four policies: random, weak-evidence-first, dense-first, and boundary-first removal. These policies model arbitrary omissions, loss of low-evidence spots, loss of crowded-region spots, and ambiguity near local cluster boundaries, respectively.

The degraded reference is denoted by $R_i^{v,\pi,\alpha}$, where v is the reference view, π is the missingness policy, and α is the retained reference fraction. We quantify the F1 drop induced by reference degradation as:

$$\Delta \mathrm{F1}_m^{v,\pi,\alpha} = \mathrm{F1}_m^{v,\pi,1.0} - \mathrm{F1}_m^{v,\pi,\alpha}. \tag{2}$$

Table 6: Structured reference-missingness policies used in annotation-retention stress tests.

Policy	Removal order	Intended annotation failure mode
Random	Uniform random removal of reference points.	Arbitrary missing annotations.
Weak-evidence-first	Removes points with weaker local evidence first.	Faint or visually ambiguous spots are more likely to be missed.
Dense-first	Removes points in locally crowded neighborhoods first.	Dense clusters are harder to annotate completely.
Boundary-first	Removes points near local cluster boundaries first.	Boundary-adjacent spots are more ambiguous to separate consistently.

For each reference condition, detectors are ranked by F1. We compare rankings obtained from degraded references with rankings obtained from the corresponding 100% reference view using Spearman rank correlation. Because benchmark use often depends on selecting the best-performing method, we also report top-1 agreement, defined as whether the detector ranked first under a degraded reference matches the detector ranked first under the corresponding 100% reference view. Detailed policy and stability-metric definitions are provided in Appendix D and Appendix E.

We also perform three robustness checks: excluding the rule-candidate detector, varying the point-matching radius, and estimating uncertainty with image-level bootstrap resampling. These analyses test whether the observed reference sensitivity is driven by candidate-reference circularity, a single matching tolerance, or sampling variability. Robustness results are reported in Table 7 and Appendix H.

4 Reference definitions alter benchmark outcomes

We first examine whether benchmark outcomes remain stable across the Raw, Lenient, and Strict reference views. Stored scored predictions, test images, and the matching protocol are held fixed across these comparisons. Prediction generation is unchanged; evaluation-score filtering follows the analysis-specific procedures in Appendix B.

As shown in Figure 2, measured F1 scores of the same detectors differ across reference views. These results show that reported performance depends on the reference annotations and evaluation-score filtering, in addition to the stored detector predictions.

Score changes become benchmark-level changes when they alter which detector would be selected

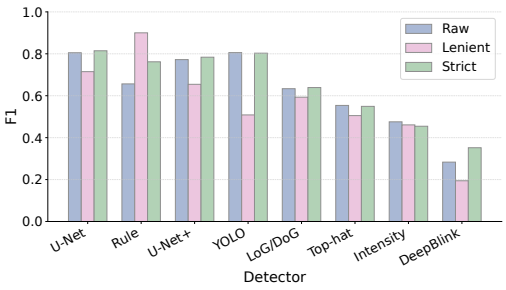


Figure 2: **Reference definitions change detector F1 scores.** F1 scores under Raw, Lenient, and Strict references, using the same stored scored predictions and matching rules.

as best. As shown in Figure 3, detector rankings differ across Raw, Lenient, and Strict under the evaluation procedures described above. Thus, these evaluation choices can affect not only absolute scores but also model-selection outcomes.



Figure 3: **Reference definitions can change model ranking.** Each cell reports the F1 score of a detector under a subset-reference condition. Although stored scored predictions and matching rules are fixed, changing the reference definition reshapes the score landscape and changes the induced model ranking.

5 Structured missingness reveals score fragility

We next test whether benchmark scores remain stable when reference annotations are missing in structured ways. Detector predictions, test images, and the matching protocol are fixed, while the retained reference fraction is reduced from 100% to 50% under Raw, Lenient, and Strict references. We consider four missingness policies: random removal, weak-evidence-first removal, dense-first removal, and boundary-first removal. These policies represent different plausible annotation failures in dense microscopy images, including arbitrary omissions, low-evidence spot omissions, crowded-region omissions, and boundary-adjacent ambiguity.

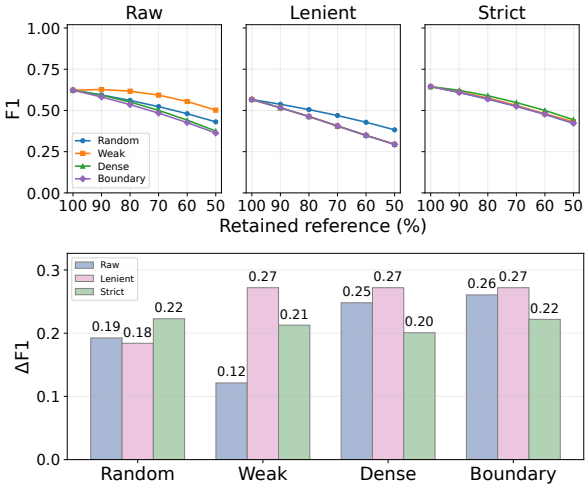


Figure 4: **Structured missingness stress tests reveal score fragility. Top:** F1 curves as reference completeness decreases from 100% to 50% under Raw, Lenient, and Strict references. **Bottom:** ΔF1 drops between 100% and 50% completeness across missingness policies. Weak, Dense, and Boundary denote weak-evidence-first, dense-first, and boundary-first removal.

Figure 4 shows that F1 changes not only with the amount of retained annotation, but also with the missingness policy. This indicates that incomplete references can bias evaluation even when detector predictions are unchanged. The same detector outputs may therefore receive different scores depending on whether missing annotations are random or concentrated among weak-evidence, dense-region, or boundary-region spots.

The policy-dependent ΔF1 results further show that annotation missingness cannot be treated as a simple reduction in label quantity. Instead, the mechanism of missingness matters. References that

lose weak, crowded, or boundary-adjacent spots can produce different degradation patterns even at the same retained-reference fraction. This supports the need for structured reference-completeness stress tests in dense microscopy spot-detection benchmarks.

6 Reference incompleteness destabilizes model selection

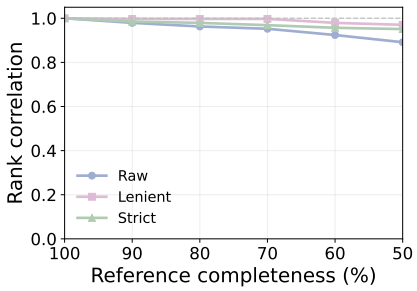


Figure 5: **Aggregate rank correlations remain relatively stable under reference missingness.** Rank correlation is computed between model rankings obtained from the corresponding 100% reference view and degraded references as reference completeness decreases. Although reference missingness perturbs F1 scores, the overall ranking order can appear relatively stable under this aggregate metric.

Figure 4 shows that structured reference missingness can perturb measured F1 scores. However, score degradation alone does not determine whether benchmark-based model selection changes, because all detectors may degrade while preserving a similar relative order. We therefore next evaluate whether reference incompleteness affects model ranking and, more importantly, the detector that would be selected as best.

We first compare the ranking induced by each degraded reference with the ranking obtained from the corresponding 100% reference view. As shown in Figure 5, rank correlations remain relatively high across reference-completeness levels. Even at 50% reference completeness, the mean rank correlation remains 0.891 for Raw, 0.970 for Lenient, and 0.951 for Strict references. This indicates that aggregate ranking metrics may suggest a degree of stability even when the reference is incomplete.

However, aggregate rank correlation does not directly measure whether the same detector would be selected as the top-performing method. This distinction is important because many benchmark conclusions ultimately depend on top-model selection rather than the full ordering of all detectors. We therefore compute top-1 agreement, defined as whether the detector ranked first under a degraded reference matches the detector ranked first under the corresponding 100% reference view.

As shown in Figure 6, top-model selection is sensitive to reference missingness under the Raw reference. At lower reference-completeness levels, degraded Raw references frequently select a different top-ranked detector, especially under random and weak-evidence-first removal. In contrast, degraded Lenient references preserve the selected top detector relative to the 100% Lenient reference view across all tested completeness levels and missingness policies. Strict references show intermediate behavior, with policy-dependent instability under weak-evidence-first, dense-first, and boundary-first removal.

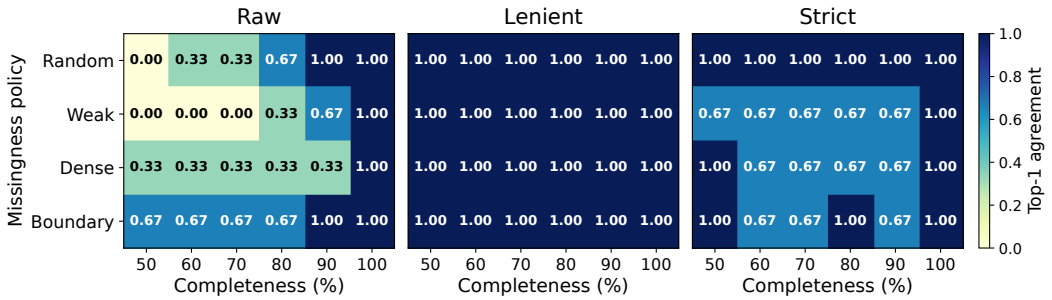


Figure 6: **Reference incompleteness can change top-model selection.** Each heatmap reports top-1 agreement across reference-completeness levels and missingness policies, averaged across evaluated subsets. A value of 1.0 means that the same detector remains top-ranked as under the corresponding 100% reference view, whereas lower values indicate top-model changes in one or more subsets. Raw references show substantial instability under missingness, while Lenient references preserve top-1 selection across all tested conditions.

Table 7: Model-selection stability with and without Rule-candidate. Top-1 agreement and rank correlation summarize the evaluated missingness conditions.

Reference	Pool	Top detector at 100%	Min. top-1 agreement	Mean rank correlation
Raw	All	U-Net	0.250	0.963
Raw	w/o Rule	U-Net	0.500	0.976
Lenient	All	Rule	1.000	0.988
Lenient	w/o Rule	U-Net	0.750	0.983
Strict	All	U-Net	0.500	0.957
Strict	w/o Rule	U-Net	0.750	0.955

Together, these results show that reference incompleteness can affect benchmark conclusions even when aggregate rank-correlation metrics appear stable. Thus, benchmark reliability should be assessed not only by score degradation or global ranking similarity, but also by the stability of top-model selection.

These results suggest an audit-guided model-selection rule. A detector should be treated as a robust benchmark winner only when its top-ranked status remains stable across plausible reference definitions and missingness conditions. When the selected top detector changes across Raw, Lenient, Strict, or degraded references, the benchmark should report reference-conditional winners rather than a single reference-independent best model.

Robustness to Rule-candidate exclusion. Because Rule-candidate shares the candidate-generation mechanism used to construct the Lenient and Strict references, we repeated the ranking-stability analysis after excluding this detector. Table 7 compares results with and without Rule-candidate.

Although removing Rule-candidate changes the baseline top detector under Lenient, top-1 agreement still decreases under incomplete-reference conditions. Thus, reference incompleteness can destabilize model selection even without Rule-candidate. Additional details are provided in Appendix H.1.

Sensitivity to reference construction. We further examined the sensitivity of the audit references to nine reference-construction settings. Although reference sizes and absolute F1 scores varied, detector rankings remained stable, with minimum Spearman correlations of 1.000 for Lenient and 0.976 for Strict relative to the default setting. Rule-candidate remained top-ranked under Lenient, while U-Net heatmap remained top-ranked under Strict across all nine settings. Detailed results are provided in Appendix F.

7 Discussion and limitations

On the three evaluated FISH_spots subsets, reference definition and completeness affect detector scores and model-selection outcomes. These findings demonstrate a concrete benchmark failure mode within the evaluated dataset.

The proposed reference views are controlled audit settings, not corrected biological ground truth. Because Lenient and Strict rely on candidate evidence, they may inherit candidate-generation bias and miss plausible faint spots. Detector-exclusion and construction-sensitivity analyses address specific sources of dependence but do not establish independence from the candidate-generation pipeline. This limitation is central to the audit framing: our goal is not to replace one imperfect reference with another, but to make reference dependence measurable.

We additionally examined local intensity evidence for Lenient-only, Raw-removed, and Raw-retained points, with representative image crops provided in Appendix G. This analysis provides image-based supporting evidence but does not replace independent expert validation.

A key implication is that score degradation and model-selection instability are related but distinct. Aggregate rank correlations can remain high while the selected top detector changes under reference missingness. This motivates reporting reference sensitivity and top-1 stability alongside standard detection metrics.

Several limitations remain. The experiments use a single public dataset, and separating sparse/near-empty images from the main audit further restricts the scope of the findings. Evaluation cutoffs, including those in the reference-invariant controls, were selected using audit annotations rather than independent calibration data. The missingness policies are simplified stress tests, not complete models of human annotation behavior. Future work should validate the framework on independent microscopy datasets with domain-expert annotations before extending the conclusions to other modalities and small-object localization tasks.

8 Conclusion

We presented a reference-audit framework for fluorescence spot-detection benchmarks. Instead of treating a single annotation set as fixed ground truth, we evaluated whether detector scores, rankings, and top-model selection remain stable under alternative reference definitions and structured missing-label mechanisms.

On the evaluated FISH_spots subsets, reference definition and completeness affected measured F1 scores, detector rankings, and top-1 model selection. After excluding Rule-candidate, reference-invariant cutoff controls showed that top-model differences between reference views persisted without reference-specific cutoff retuning.

More broadly, model rankings should be interpreted as conditional on the reference construction protocol, rather than as reference-independent properties of detectors. Raw, Lenient, and Strict references are therefore not intended to define a single definitive biological ground truth. Instead, they serve as controlled audit references for testing whether benchmark outcomes remain robust under plausible annotation ambiguity and incompleteness.

These findings suggest that fluorescence spot-detection benchmarks should report reference sensitivity, structured missingness robustness, and top-1 stability alongside standard detector performance.

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A Audit subset selection and sampling

Subset selection. We identified 550, 1,052, and 451 valid 2D image-annotation pairs in MERFISH, RCA, and SeqFISH, respectively. These subsets provide contrasting experimental imaging conditions within FISH_spots. We selected an equal number of audit images from each subset rather than sampling in proportion to subset size.

Sampling and replacement. We first randomly sampled 50 images per subset using seed 42. Within each sample, images whose Raw spot counts were at or below the subset-specific 10th percentile were assigned to a separate sparse/near-empty control set, yielding five images per subset.

Replacement images were randomly sampled from the same subset, again using seed 42. The replacement pool excluded all images in the original audit sample and images whose Raw spot counts were at or below the 10th percentile computed over all valid images in that subset. Five replacements per subset restored the main audit set to 50 images per subset, for a total of 150 images.

Split handling and scope. Replacement images were removed from their original training or validation manifests when constructing the refined splits. Table 1 reports the split sizes before this removal; the refined manifests contain five fewer training images in each of MERFISH, RCA, and SeqFISH, with validation counts unchanged.

The same final 150 audit images are used throughout the reference-view comparisons and annotation-retention stress tests. No additional audit images were generated through augmentation or synthesis. Because sparse/near-empty images were separated from the main audit, the reported results characterize the selected audit images rather than the full image distribution of FISH_spots.

B Detector settings and cutoff selection

Checkpoint selection. Learning-based detector checkpoints were selected using validation data where applicable. U-Net heatmap and U-Net heatmap-pos used checkpoints with the lowest validation loss. YOLO used the `best.pt` checkpoint saved according to the Ultralytics validation metric. DeepBlink used the official pretrained `smfish1` model without additional checkpoint selection. Checkpoint selection is distinct from evaluation-cutoff selection.

Prediction generation and post-processing. The following detector-specific settings were used to generate standardized point predictions:

- **U-Net / U-Net+POS:** Local maxima with an initial heatmap cutoff of 0.25, a minimum distance of 2 pixels, and a cap of 3,000 points per image.
- **YOLO:** Confidence floor of 0.001, non-maximum suppression IoU threshold of 0.5, and `max_det=3000`. Bounding-box centers were used as point predictions.
- **DeepBlink:** Official point predictions from the pretrained `smfish1` model, without additional spatial post-processing.
- **Intensity / Top-hat:** Local maxima above the 97.5th-percentile threshold, a minimum distance of 3 pixels, and a cap of 1,800 points per image. Top-hat used a 15×15 morphological kernel.
- **LoG/DoG:** Scales of 1.0, 1.5, and 2.0 pixels, an SNR threshold of 3.5, a 97th-percentile LoG-response threshold, a minimum distance of 2 pixels, and candidate caps of 2,500 points for MERFISH and 2,000 points for RCA and SeqFISH.
- **Rule-candidate:** Constructed candidate locations used directly, with per-subset SNR-based score normalization and no additional spatial point extraction.

These prediction-generation settings were manually prespecified and applied unchanged across reference views. They were not adjusted according to F1 under the Raw, Lenient, or Strict references or through visual inspection of the audit test images. The resulting scored predictions were stored and reused for subsequent evaluations.

Evaluation cutoff selection. For the reference-sensitivity analysis, predictions with scores below the evaluation cutoff were discarded. Cutoffs were selected from the prespecified grid $\{0, 0.05, \ldots, 0.95, 0.975, 0.99\}$.

Subset-level cutoffs maximized F1 separately for each detector, reference, and subset, using counts summed over the 50 images in that subset. The aggregate threshold sweep separately selected one cutoff per detector and reference by maximizing F1 from counts pooled across all 150 images, applying each candidate cutoff uniformly across subsets. Ties in computed F1 were resolved in favor of the lower cutoff. These selections used audit annotations rather than independent validation data.

For the Raw and Strict missingness experiments, the cutoff selected on the corresponding full reference for each detector and subset was held fixed as annotations were removed. Aggregate missingness scores were computed by pooling the resulting subset-level TP, FP, and FN counts, rather than averaging subset-level F1 scores. The Lenient missingness implementation used the stored point predictions without an additional evaluation-score cutoff.

Reference-invariant cutoff control. Using aggregate F1 across the 150 audit images, we selected one cutoff per detector and applied it unchanged across Raw, Lenient, and Strict. The Raw-selected rule maximized aggregate F1 under Raw. The pooled rule maximized the arithmetic mean of aggregate F1 across the three reference views. Both rules used the same prespecified cutoff grid. The two rules could select different numerical cutoffs, but each rule held a detector's cutoff fixed across all three reference views.

After excluding Rule-candidate, both rules selected YOLO as the top detector under Raw and Lenient and U-Net heatmap under Strict. Thus, reference-dependent top-model selection persisted without per-reference cutoff retuning. Both rules still used audit annotations and therefore do not constitute independent validation-based calibration.

Fixed and varied components. Table 8 summarizes the experimental design of the main audit comparisons and controls.

Table 8: Fixed and varied components in the main audit comparisons and controls. Evaluation scope refers to subset-level or aggregate evaluation.

Experiment	Fixed components	Varied components
Reference-specific cutoff analysis	Audit images, scored predictions, prediction-generation settings, and matching tolerance	Reference view; evaluation cutoff selected separately for each detector and evaluation scope
Missingness stress tests	Audit images, scored predictions, matching tolerance, and prediction filtering within each reference/subset	Reference view, retained fraction, and removal policy
Reference-construction sensitivity	Audit images, scored predictions, evaluation cutoffs, and matching tolerance	Reference-construction parameters
Reference-invariant cutoff control	Audit images, scored predictions, matching tolerance, and one cutoff per detector within each selection rule	Reference view

C Additional evaluation details

For image I_i, detector m, and reference view v, let $P_{i,m}$ denote the predicted point set, R_i^v denote the reference point set, and $M_{i,m}^v$ denote the one-to-one matched pair set. Predictions and reference points are matched within the feasible edge set

$$E_{i,m}^v = \{(p, r) \in P_{i,m} \times R_i^v : d(p, r) \leq \tau_{\text{eval}}\}. \tag{3}$$

We compute a one-to-one matched set $M_{i,m}^v \subseteq E_{i,m}^v$ such that each prediction and each reference point is used at most once. When multiple feasible assignments exist, matches are assigned to prioritize shorter prediction-reference distances.

The image-level counts are

$$\text{TP}_{i,m}^v = |M_{i,m}^v|, \quad \text{FP}_{i,m}^v = |P_{i,m}| - |M_{i,m}^v|, \quad \text{FN}_{i,m}^v = |R_i^v| - |M_{i,m}^v|. \tag{4}$$

Dataset-level counts are aggregated over images:

$$\text{TP}_m^v = \sum_{i=1}^{N} \text{TP}_{i,m}^v, \quad \text{FP}_m^v = \sum_{i=1}^{N} \text{FP}_{i,m}^v, \quad \text{FN}_m^v = \sum_{i=1}^{N} \text{FN}_{i,m}^v. \tag{5}$$

Precision, recall, and F1 score are defined as

$$\text{Precision}_m^v = \frac{\text{TP}_m^v}{\text{TP}_m^v + \text{FP}_m^v}, \quad \text{Recall}_m^v = \frac{\text{TP}_m^v}{\text{TP}_m^v + \text{FN}_m^v}, \tag{6}$$

$$\text{F1}_m^v = \frac{2 \cdot \text{Precision}_m^v \cdot \text{Recall}_m^v}{\text{Precision}_m^v + \text{Recall}_m^v}. \tag{7}$$

D Missingness policy details

For reference view v, missingness policy π, and retained fraction $\alpha \in (0, 1]$, the degraded reference is denoted by $R_i^{v,\pi,\alpha} \subseteq R_i^v$ and satisfies

$$|R_i^{v,\pi,\alpha}| = \lfloor \alpha |R_i^v| \rfloor . \tag{8}$$

Each missingness policy defines a removal-priority function $\phi_\pi(r)$ for reference point r. Points with higher $\phi_\pi(r)$ are removed earlier. Equivalently, the retained degraded reference is obtained by keeping the points with the lowest removal priorities:

$$R_i^{v,\pi,\alpha} = \mathrm{BottomK}_{r \in R_i^v} \left(\phi_\pi(r), \lfloor \alpha |R_i^v| \rfloor \right) . \tag{9}$$

Random removal assigns independent random priorities:

$$\phi_{\mathrm{random}}(r) \sim \mathrm{Uniform}(0, 1). \tag{10}$$

Weak-evidence-first removal prioritizes points with weaker local evidence:

$$\phi_{\mathrm{weak}}(r) = -q_i(r), \tag{11}$$

where $q_i(r)$ is a local evidence score derived from candidate-support statistics and image-level signal evidence. Thus, lower-evidence points receive higher removal priority and are removed earlier.

Dense-first removal prioritizes points in locally crowded neighborhoods. We define local point density as

$$\rho_i(r) = \sum_{r' \in R_i^v \setminus \{r\}} \mathbb{I} \left[d(r, r') \le \epsilon_{\mathrm{dens}} \right] , \tag{12}$$

and set

$$\phi_{\mathrm{dense}}(r) = \rho_i(r). \tag{13}$$

Boundary-first removal prioritizes points near local cluster boundaries:

$$\phi_{\mathrm{boundary}}(r) = \frac{1}{\delta_i(r) + \epsilon}, \tag{14}$$

where $\delta_i(r)$ denotes the distance from point r to the estimated boundary of its local spot cluster, and ϵ is a small constant for numerical stability.

E Ranking and top-1 metric details

Let $\mathcal{M}$ denote the set of evaluated detectors. For each reference condition (v, π, α), detector rankings are induced by F1:

$$\mathrm{rank}_{v,\pi,\alpha}(m) = 1 + \sum_{m' \in \mathcal{M} \setminus \{m\}} \mathbb{I} \left[\mathrm{F1}_{m'}^{v,\pi,\alpha} > \mathrm{F1}_{m}^{v,\pi,\alpha} \right] . \tag{15}$$

We compare the ranking under degraded references with the ranking under the corresponding 100% reference view using Spearman rank correlation:

$$\rho_{\mathrm{rank}}^{v,\pi,\alpha} = \mathrm{corr}_{\mathrm{Spearman}} \left(\{\mathrm{rank}_{v,\pi,\alpha}(m)\}_{m \in \mathcal{M}}, \{\mathrm{rank}_{v,\pi,1.0}(m)\}_{m \in \mathcal{M}} \right) . \tag{16}$$

The top-ranked detector under condition (v, π, α) is

$$m_{v,\pi,\alpha}^\star = \arg\max_{m \in \mathcal{M}} \mathrm{F1}_m^{v,\pi,\alpha}. \tag{17}$$

Top-1 agreement is defined as

$$A_{\mathrm{top1}}^{v,\pi,\alpha} = \mathbb{I} \left[m_{v,\pi,\alpha}^\star = m_{v,\pi,1.0}^\star \right] . \tag{18}$$

When results are averaged across subsets $\mathcal{S}$, top-1 agreement is computed as

$$\bar{A}_{\mathrm{top1}}^{v,\pi,\alpha} = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \mathbb{I} \left[m_{s,v,\pi,\alpha}^\star = m_{s,v,\pi,1.0}^\star \right] . \tag{19}$$

F Reference-construction sensitivity

We examined the dependence of the audit results on the parameters used to construct the Lenient and Strict reference views. Four parameters were varied individually around their default settings while keeping the remaining parameters fixed, yielding nine unique settings in total: one default setting and two perturbations for each parameter.

Table 9: Parameter variations used in the reference-construction sensitivity analysis. Only one parameter was varied at a time.

Parameter	Evaluated settings
Reference-matching tolerance	2, 3 (default), and 5 pixels
SNR threshold	2.5, 3.0 (default), and 3.5
LoG-response percentile	Subset-specific default and shifts of ± 2 percentile points
Candidate cap	$0.5\times$, $1\times$ (default), and $1.5\times$ the default cap

The same audit images and stored detector predictions were used across all settings. Detector-specific evaluation cutoffs and the evaluation-matching tolerance $\tau_{\text{eval}} = 3$ pixels were held fixed.

Across the nine settings, Lenient reference sizes ranged from 61,135 to 141,204 points, while Strict reference sizes ranged from 43,902 to 63,514 points. The largest detector-specific F1 range was 0.253 under Lenient and 0.120 under Strict. The LoG-response threshold had the largest effect on reference size and absolute F1, whereas the matching tolerance, SNR threshold, and candidate cap had comparatively smaller effects.

Despite these variations, detector rankings remained stable relative to the default setting. The minimum Spearman rank correlation was 1.000 under Lenient and 0.976 under Strict. With all eight detectors included, Rule-candidate ranked first under Lenient and U-Net heatmap ranked first under Strict in all nine settings.

These results indicate that absolute reference sizes and F1 scores depend on construction parameters, while the observed difference in top-model selection between Lenient and Strict persists across the tested settings.

G Image-evidence analysis of audit reference points

We examined local image evidence for three point categories: Lenient-only points, Raw points removed under Strict, and Raw points retained under Strict.

Image intensities were normalized to $[0, 1]$. For each point, local peak intensity was defined as the maximum intensity within a 5×5 pixel neighborhood. Local contrast was computed as the difference between this peak and the median intensity in a 3-5 pixel annulus surrounding the point.

Table 10 summarizes the local intensity and contrast statistics.

Table 10: Local image-evidence statistics for the three audit point categories. Positive contrast denotes a local peak above the surrounding background.

Point category	n	Median peak	Median contrast	Positive contrast
Lenient-only	43,742	0.600	0.400	99.97%
Raw removed	22,466	0.835	0.402	97.99%
Raw retained	57,408	0.906	0.518	100.00%

Points in all three categories generally coincide with localized intensity above the surrounding background. Raw-retained points exhibit the strongest image evidence, whereas Lenient-only and Raw-removed points show lower local contrast.

The intensity analysis and representative crops provide supporting image-based evidence, but they do not establish biological ground truth or substitute for independent assessment by domain experts.

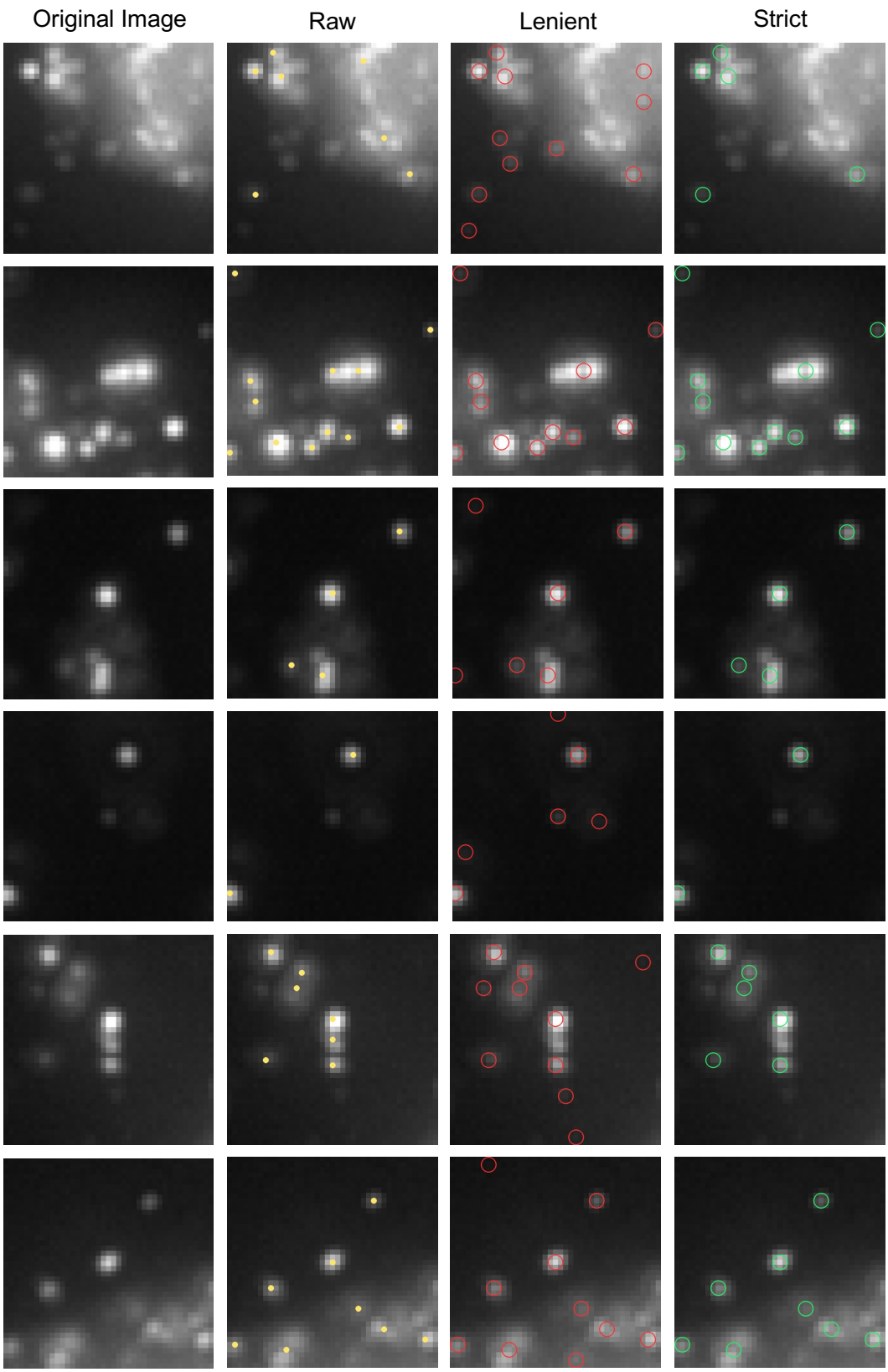


Figure 7: **Local comparison of audit reference views.** Six selected 41 × 41-pixel crops from MERFISH_179. Columns: original image, Raw (yellow dots), Lenient (red circles), and Strict (green circles). Examples illustrate added, excluded, and retained points (two regions each) and are qualitative, not randomly sampled.

H Robustness checks

We perform three robustness checks to test whether the observed reference sensitivity depends on specific implementation choices: excluding the rule-candidate detector, varying the point-matching radius, and estimating uncertainty with image-level bootstrap resampling.

H.1 Excluding the Rule-candidate detector

Because candidate evidence is used to construct the Lenient and Strict reference views, we repeated ranking and top-1 stability analyses after excluding Rule-candidate from the detector pool.

The results are summarized in Table 7. Reference-dependent model-selection instability remains observable after excluding Rule-candidate, indicating that the findings are not solely attributable to this detector.

H.2 Sensitivity to point-matching radius

The main experiments use $\tau_{\text{eval}} = 3$ pixels. To test whether the conclusions depend on this single tolerance, we repeat the reference-sensitivity and top-1 stability analyses using alternative matching radii.

Table 11: Sensitivity of reference-dependent conclusions to the point-matching radius. Although absolute F1 scores vary with the matching radius, the main conclusion that reference definition and reference incompleteness can alter model selection remains consistent.

τ_{eval}	Reference view	Top detector at 100%	Min. top-1 agreement	Mean rank correlation
2 px	Raw	U-Net heatmap	0.250	0.963
2 px	Lenient	Rule-candidate	1.000	0.979
2 px	Strict	U-Net heatmap	0.500	0.964
3 px	Raw	U-Net heatmap	0.250	0.963
3 px	Lenient	Rule-candidate	1.000	0.988
3 px	Strict	U-Net heatmap	0.500	0.957
5 px	Raw	U-Net heatmap	0.000	0.956
5 px	Lenient	Rule-candidate	1.000	0.968
5 px	Strict	U-Net heatmap	0.750	0.960

The same qualitative pattern appears across all tested matching radii. Rank correlations remain high, but top-1 agreement can still be low, especially under Raw references. This supports the conclusion that aggregate rank-correlation metrics can mask instability in top-1 model selection.

H.3 Image-level bootstrap uncertainty

We estimate uncertainty using image-level bootstrap resampling over the held-out audit images. For each bootstrap sample, we recompute F1, detector rankings, rank correlation, and top-1 agreement. This analysis tests whether the observed model-selection instability is driven by a small number of images.

Table 12: Bootstrap uncertainty of top-1 detector selection. Each entry reports the bootstrap selection probability of a detector being ranked first under the corresponding reference view. A dash indicates that no additional detector obtained a nonzero top-1 selection probability.

Reference view	Detector 1	Detector 2	Detector 3
Raw	U-Net heatmap (0.887)	YOLO point-box (0.113)	–
Lenient	Rule-candidate (1.000)	–	–
Strict	U-Net heatmap (1.000)	–	–

The bootstrap analysis shows that Raw references retain moderate top-1 uncertainty, with YOLO point-box selected as the top detector in 11.3% of bootstrap samples. Lenient and Strict references

yield concentrated top-1 selections within each reference view. However, the identity of the selected top detector differs across reference definitions, indicating that reference-dependent model-selection changes cannot be explained solely by image-level sampling variability.

I Additional subset-level missingness results

We provide subset-level missingness stress-test results for MERFISH, RCA, and SeqFISH. These plots complement the aggregate results in Figure 4 by showing how F1 curves change within each evaluated subset. Across subsets, F1 varies with both the retained reference fraction and the missingness policy, indicating that reference incompleteness is not only a matter of annotation quantity but also of which annotations are missing.

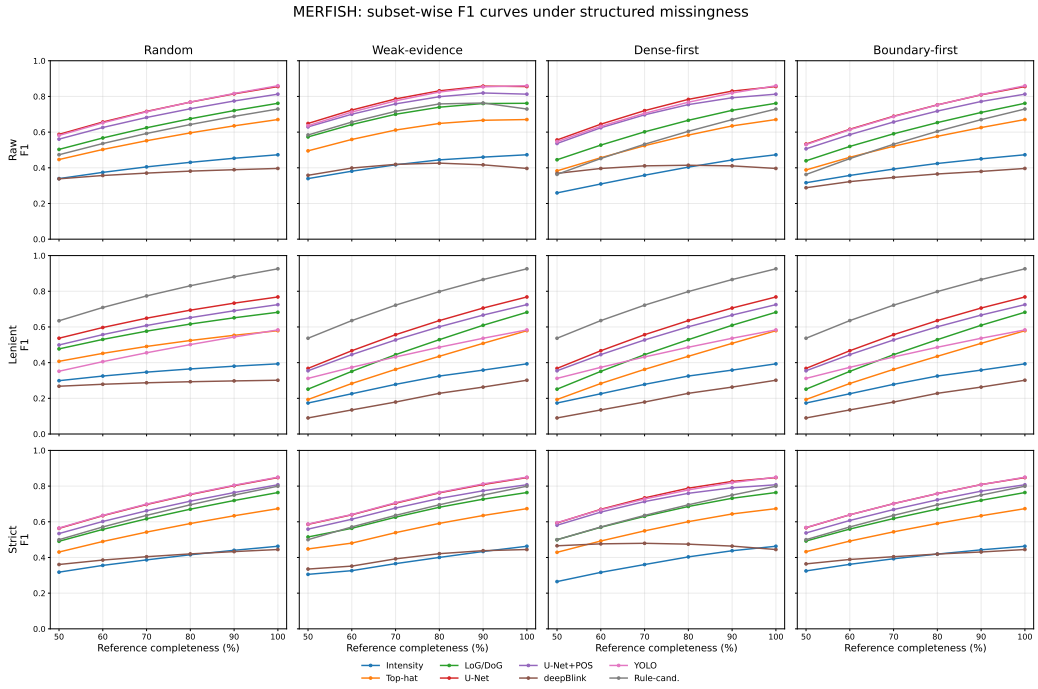


Figure 8: **Subset-level structured missingness results for MERFISH.** Detector predictions are fixed while the retained reference fraction decreases from 100% to 50% under Raw, Lenient, and Strict references. F1 is recomputed under random, weak-evidence-first, dense-first, and boundary-first missingness policies.

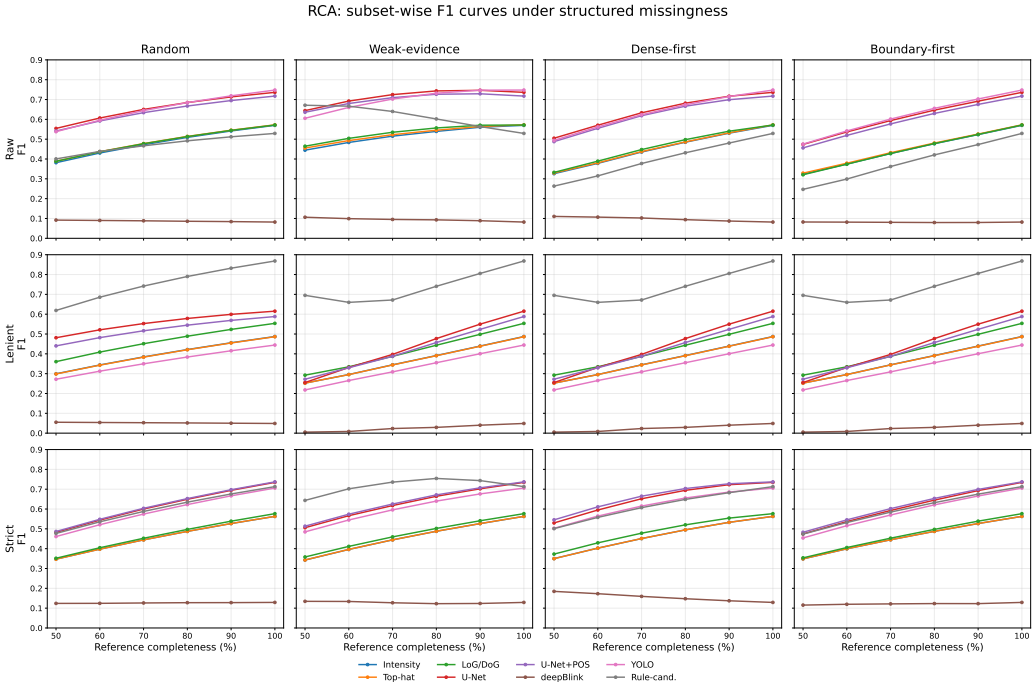


Figure 9: **Subset-level structured missingness results for RCA.** Detector predictions are fixed while the retained reference fraction decreases from 100% to 50% under Raw, Lenient, and Strict references. F1 is recomputed under random, weak-evidence-first, dense-first, and boundary-first missingness policies.

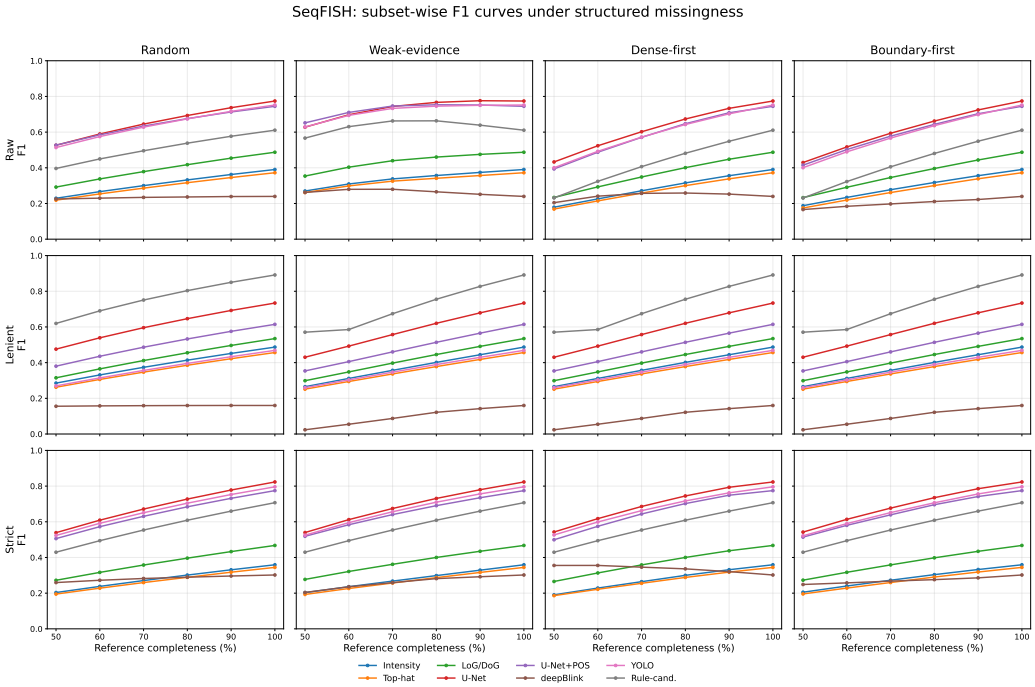


Figure 10: **Subset-level structured missingness results for SeqFISH.** Detector predictions are fixed while the retained reference fraction decreases from 100% to 50% under Raw, Lenient, and Strict references. F1 is recomputed under random, weak-evidence-first, dense-first, and boundary-first missingness policies.

J Additional ranking-stability visualizations

We provide subset-level Spearman rank-correlation heatmaps for each reference view. These visualizations complement Figure 5 by showing ranking stability across reference views and missingness policies within each subset. Higher values indicate stronger agreement with the detector ranking obtained from the corresponding 100% reference view. The subset-level heatmaps show that rank correlation often remains high, particularly under Lenient references, but can drop substantially for specific subset-reference-policy combinations such as RCA under Raw weak-evidence removal. Together with the top-1 agreement analysis, these results show that aggregate rank correlation alone is insufficient for assessing benchmark reliability.

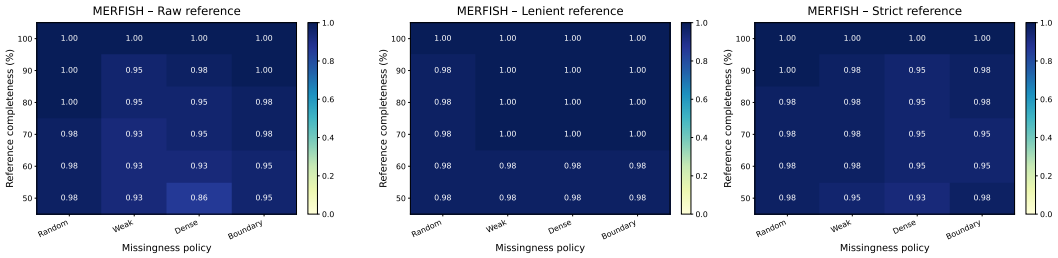


Figure 11: **Subset-level rank-stability heatmaps for MERFISH.** Left to right: Raw, Lenient, and Strict reference views. Each cell reports the Spearman rank correlation between detector rankings under degraded references and the corresponding 100% reference view.

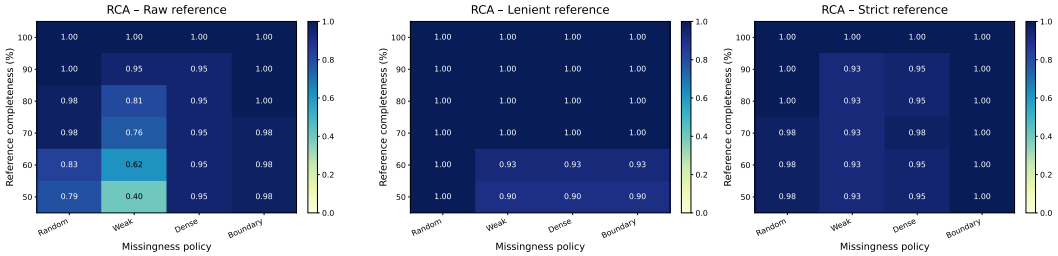


Figure 12: **Subset-level rank-stability heatmaps for RCA.** Left to right: Raw, Lenient, and Strict reference views. RCA under Raw weak-evidence removal shows a pronounced drop in rank correlation at lower completeness levels, illustrating that subset-specific instability can be stronger than aggregate trends suggest.

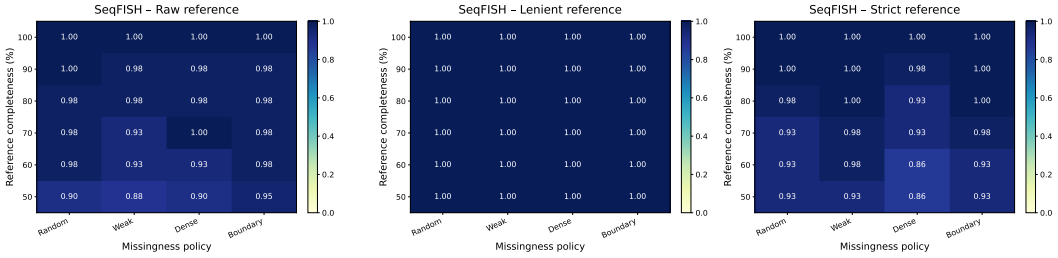


Figure 13: **Subset-level rank-stability heatmaps for SeqFISH.** Left to right: Raw, Lenient, and Strict reference views. Each cell reports the Spearman rank correlation between detector rankings under degraded references and the corresponding 100% reference view.

K Compute resources

Reference construction, point-matching evaluation, missingness stress tests, ranking analysis, and figure generation were lightweight and can be run on standard CPU hardware. Learning-based

detector training and inference were performed on a local workstation with four NVIDIA RTX 6000 Ada GPUs. The public code repository includes the software environment and commands required to reproduce the main audit results.

L Existing assets and data access

This study uses the public FISH_spots dataset as an existing asset [17, 16]. The original dataset is hosted at: `https://huggingface.co/datasets/GangCaoLab/FISH_spots`.

Raw microscopy images are not redistributed as standalone files in the audit package. Users should obtain them from the original hosting page and follow the applicable source-data terms. The package's `DATA.md` describes the expected directory structure and the `FISH_SPOTS_ROOT` configuration.

The audit package contains the image-level audit manifest, standardized reference coordinates, stored detector predictions, and evaluation outputs used in the reported analyses. The reference files used for the main audit contain 79,874 Raw points, 101,150 Lenient points, and 57,408 Strict points across the same 150 images.

M Code availability and reproducibility

The code, environment specifications, processed audit references, standardized detector predictions, evaluation results, and figure-generation scripts are publicly available at: `https://github.com/jiyoungzzv/fish_reference_audit`.

Provided artifacts. The audit package includes:

- Image-level audit manifests and split summaries in `data_manifest/`.
- Standardized Raw, Lenient, and Strict reference CSVs in `results/references/standardized/`.
- Stored scored detector predictions in `results/predictions/standardized/`.
- Evaluation scripts, precomputed result CSVs, and scripts for generating the main quantitative plots.
- Environment specifications in `environment.yml` and `requirements.txt`, with optional inference dependencies in `requirements-optional.txt`.

Reproduction workflow. The package is organized around audit computations from standardized references and stored detector predictions. This workflow separates evaluation from detector training and inference. The `README.md` and `REPRODUCE.md` document commands for reference-sensitivity evaluation, annotation-retention stress tests, ranking analysis, and figure generation.

The plotting scripts for Figures 2 and 3, Figure 4, Figure 5, and Figure 6 use the corresponding stored evaluation outputs. Regenerating these plots from the stored outputs does not require detector retraining or inference. This scope is distinct from end-to-end reproduction starting with raw images and detector training.

Random seeds. Audit split construction uses seed 42. The Raw and Strict random-removal implementation uses a base seed of 42 with run indices 0-19. The separate Lenient random-removal implementation uses seeds 0-19 directly. These implementation-specific seed conventions are recorded in the reference-generation scripts.

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   Answer: [Yes]

   Justification: The abstract and introduction state the paper's scope as a benchmark-audit study rather than a new detector proposal. The main claims are supported by the reference-sensitivity experiments, structured missingness stress tests, and model-selection stability analyses reported in Sections 3–6.

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